

LLD-GS-001 GROUND SEGMENT & DATAFLOW ENDPOINT

Low Level Design — Ground Segment & Dataflow Endpoint

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VITA 49 | CCSDS TM/TC | ECS Fargate | Direct Connect | AWS Arc
SpaceNet-IT | HLD Ref: SPACENET-HLD-001 | June 2026

1. Scope & Purpose

This LLD defines the detailed design of the Ground Segment and Dataflow Endpoint tier. It covers the AWS Ground Station managed antenna configuration, hybrid on-premises connectivity, VITA 49 UDP dataflow endpoint container, CCSDS TM frame decoder, and the initial routing of decoded telemetry into the AWS data pipeline.

2. AWS Ground Station Configuration

2.1 Antenna Locations

Location	Region	Frequency Bands	Primary Coverage
Sydney, Australia	ap-southeast-2	S-band / X-band	Primary — LEO passes over Asia-Pacific
Hawaii, USA	us-west-2	S-band / X-band	Pacific rim coverage gap fill
Oregon, USA	us-west-2	S-band	North America passes
Dublin, Ireland	eu-west-1	S-band / X-band	Europe / North Atlantic passes
Frankfurt, Germany	eu-central-1	S-band	Central Europe passes
São Paulo, Brazil	sa-east-1	S-band	South America passes
Singapore	ap-southeast-1	S-band / X-band	South-East Asia passes
Cape Town, S. Africa	af-south-1	S-band	Africa / Indian Ocean passes

2.2 Contact Profile Design

Parameter	Configuration	Notes
Antenna azimuth rate	3°/sec max	Standard LEO tracking
Elevation mask	10° minimum	Avoid atmospheric distortion
Autotrack mode	Enabled	Closed-loop RF tracking
Polarisation	RHCP primary	LHCP fallback configurable
S-band uplink freq	2025–2110 MHz	CCSDS TC uplink band
X-band downlink freq	8025–8400 MHz	Payload data downlink
S-band downlink freq	2200–2290 MHz	TT&C housekeeping downlink
Demodulation	BPSK / QPSK / 8PSK	Software-defined radio
FEC	CCSDS concatenated (RS + Conv) / Rate 1/2 Viterbi + RS(255,223)	
Data delivery	UDP to dataflow endpoint	VITA 49 VRT packets

2.3 Dataflow Endpoint Group

Endpoint type	Customer dataflow endpoint (EC2 / ECS in VPC)
UDP port	55888 (AWS Ground Station default)
Endpoint IP	Private IP in ground-segment-prod VPC (10.10.2.10)
Max throughput	200 Mbps per contact
Delivery format	VITA 49.2 (VRT) packets
Security group	Inbound 55888/UDP from AWS Ground Station service CIDR only

3. VITA 49 Dataflow Endpoint — Detailed Design

3.1 Packet Structure

Packet Type	4 bits	0001=Signal Data, 0100=Context, 0110=Command
Class ID Present	1 bit	Set 0 in AWS Ground Station delivery
Trailer Present	1 bit	Set 0 in standard delivery
TSI (Timestamp Int)	2 bits	01 = UTC seconds
TSF (Timestamp Frac)	2 bits	10 = Real-time (picoseconds)
Packet Count	4 bits	Rolling mod-16 sequence counter
Packet Size	16 bits	Total packet size in 32-bit words
Stream ID	32 bits	Unique per satellite contact stream
Integer Timestamp	32 bits	UTC seconds since J2000
Fractional Timestamp	64 bits	Picoseconds within current second
Payload	Variable	I/Q samples (16-bit signed pairs) or JSON context

3.2 ECS Fargate Container Design

Container image	spacenet/dataflow-endpoint:latest (ECR private)
Task CPU	2 vCPU
Task Memory	4 GB
UDP bind	0.0.0.0:55888
Receive buffer	65536 bytes (max UDP datagram)

Internal queue	10,000 frame deep deque
Processing threads	4 (decoder workers)
Health check	HTTP /health on port 8080
Scaling	Target tracking — CPU 60% threshold
IAM task role	sts:AssumeRole → Kinesis PutRecord, IoT Publish, S3 PutObject
Logging	CloudWatch Logs /spacenet/dataflow-endpoint

3.3 Signal Processing Pipeline (per packet)

- Receive UDP datagram on port 55888
- Parse VITA 49 header — extract packet type, stream ID, timestamp
- Context packet → update RF metadata cache (frequency, bandwidth, gain, satellite ID)
- Signal data packet → extract I/Q payload bytes
- Pass I/Q bytes to CCSDS TM frame decoder (see Section 4)
- On decode success → route to Kinesis + IoT Core + S3
- On CRC failure → increment error counter, send CloudWatch metric, discard frame
- Contact end detection → flush queue, publish contact summary to Kinesis events stream

4. CCSDS TM Frame Decoder — Detailed Design

4.1 Frame Format (CCSDS 132.0-B-2)

Primary Header	6	SCID, VCID, frame counts, OCF flag, data field status
Secondary Header	10	2-byte extension ID + 8-byte CCSDS CDS timestamp
Data Field	1095	Space Packets — variable, padded with 0xE0 idle fill
Frame Error Control	2	CRC-16/CCITT over entire frame except FECF
Total Frame Length	1113	Fixed length — standard CCSDS TM frame

4.2 Virtual Channel Routing

0	Housekeeping	0x100–0x1FF	HK-{SCID}	spacenet-telemetry-hk stream
1	Stored TM	0x200–0x2FF	STORED-{SCID}	spacenet-telemetry-hk stream (replay flag)
2	Science Data	0x300–0x3FF	SCI-{SCID}	spacenet-telemetry-science stream
3	Fill/Idle	0x7FF	Discarded	Dropped — no routing

4.3 Space Packet Decoder (CCSDS 133.0-B-2)

0x100	Power Subsystem	bus_voltage, bus_current, battery_soc, solar_power, eclipse_flag
0x101	Thermal Subsystem	obc_temp, battery_temp, solar_panel_temp, rx_temp
0x102	ADCS	roll, pitch, yaw, roll_rate, pitch_rate, mode
0x103	OBC	cpu_load, mem_used, uptime, mode, last_cmd
0x104	Communications	rss_i, tx_power, tx_freq, rx_freq, link_quality
0x105	Orbital State	altitude, velocity, latitude, longitude

5. Hybrid On-Premises Ground Segment

5.1 Direct Connect Architecture

Connection type	AWS Direct Connect Dedicated (10Gbps)	Site-to-Site VPN (BGP, 2 tunnels)
Physical location	On-premises MDF → AWS Direct Connect PoP	Internet breakout → AWS VGW
BGP ASN	65001 (on-prem) ↔ 64512 (AWS)	65001 (on-prem) ↔ 64512 (AWS)
Route preference	BGP local pref 200 (preferred)	BGP local pref 100 (backup)
Failover time	< 30 seconds (BFD detection)	< 60 seconds (IKEv2 rekey)
MTU	9001 (jumbo frames)	1500 (standard)

5.2 AWS Arc Configuration

Ground station servers	Arc-enabled servers (Connected Machine Agent)	Unified patch, policy, monitoring
Antenna controllers	Arc-enabled Linux — read-only Arc	Health monitoring via Azure Monitor Agent
Stack HCI nodes	Arc-enabled HCI cluster	Local TC processing, S3 DataSync source
Network equipment	Not Arc-enabled (monitoring via SNMP/CloudWatch)	Health telemetry

6. Error Handling & Resilience

UDP packet loss (RF noise)	VITA49 sequence gap	Log + discard, increment metric	FEC corrects < 10% BER
CCSDS CRC failure	CRC mismatch	Discard frame, CloudWatch alarm	Retransmit not possible (downlink only)

ECS task crash	ECS health check fails	ECS task replacement (< 30s)	SQS dead-letter for in-flight packets
Kinesis shard throttle	ProvisionedThroughputExceeded	Exponential backoff retry (KPL)	Auto-scaling adds shards
Direct Connect outage	BGP session drop	VPN failover (BFD < 30s)	Direct Connect restores automatically
Ground Station contact failure	No UDP packets received	Alert NOC, try secondary region	Re-schedule contact at next window

7. Security Controls

Network isolation	VPC with no IGW on processing subnets, SG whitelist UDP/55000-55010, GS CIDR only	NIST SC-7
Encryption in transit	TLS 1.3 on all API calls; UDP in VPC (encrypted VPC traffic)	NIST SC-8
Encryption at rest	KMS CMK on S3 raw archive, DynamoDB, Kinesis	NIST SC-28
IAM task role	Least privilege — PutRecord Kinesis, PutObject S3, Publish IoT MQTT	NIST AC-6
Container image scanning	ECR image scan on push — block CRITICAL findings	NIST SI-3
Secrets management	AWS Secrets Manager — no hardcoded credentials in container	NIST IA-5
Audit logging	CloudTrail + VPC Flow Logs + ECS container logs → CloudWatch	NIST AU-2

8. Monitoring & Alerting

UDP packets/sec	Custom CloudWatch metric	< 1000 pps during contact	SNS alert to NOC
CCSDS CRC error rate	Lambda custom metric	> 5% error rate	CloudWatch alarm → NOC page
ECS task CPU	CloudWatch Container Insights	> 80% 5min avg	ECS scale-out policy
Kinesis PutRecord latency	CloudWatch	> 500ms p99	CloudWatch alarm
Direct Connect state	CloudWatch DX metrics	StateDown	Critical alert — failover
Contact data bytes	S3 PutObject size	0 bytes after contact	Contact failure alarm